CAN AI HELP A FOUNDRY ENGINEER?

A Practical Evaluation of AI Assistants for Casting Defects and Shop-floor Problem Solving

A Personal Learning Project

Prepared by:
Piyush Ranjan
B.Tech, Metallurgy and Materials Engineering
National Institute of Advanced Manufacturing Technology (NIAMT), Ranchi

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I want to thank the senior engineers and operators at my workplace who, without knowing it, taught me most of what I used to judge whether these AI answers were actually correct. A lot of the "personal observation" comments in this report come directly from something I was told on the shop floor at some point, not from a book. I also want to thank the AI Assistant for being patient with a fairly large number of repeated and sometimes badly worded questions over a couple of weeks.

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1. Introduction

I graduated in Metallurgy and Materials Engineering and joined the shop floor of a manufacturing plant soon after. Like most fresh graduate engineer trainees, I found that a lot of the practical, day-to-day questions I had didn't really match what I'd studied in college - things like why a particular casting failed a hydro test, or why the spectrometer gave two slightly different readings on the same sample.

Around the same time, I noticed a lot of people, including some of my own batchmates, casually using an AI assistant to ask technical questions. That made me curious about something specific: is this actually reliable enough for a fresh foundry engineer to use, or is it just good at sounding confident?

So I decided to actually test it properly instead of just assuming an answer either way. I picked 20 real, practical questions from foundry engineering - the kind I've either asked myself or heard being asked around me - and evaluated the AI's answers the same way I would evaluate an answer from a fellow trainee: is it correct, is it complete, is it clear, and would it actually help me on the floor.

This report is the result of that exercise.

2. Objectives

- To check whether a general AI Assistant can answer practical foundry engineering questions correctly.
- To evaluate the answers from the point of view of a Graduate Engineer Trainee, not an expert.
- To find out in which topics AI performs well and in which topics it struggles.
- To understand the practical limitations of using AI for shop-floor problem solving.
- To form my own honest opinion on whether this is something worth recommending to other trainees.

3. Methodology

I used the free version of ChatGPT for this exercise, since that's what's actually accessible to most trainees, and I wanted the evaluation to reflect what a real GET would experience, not a paid or specially configured version. For the rest of this report, I've just referred to it as the "AI Assistant" rather than repeating the product name everywhere.

I collected 20 questions across six practical areas of foundry work: casting defects (including green sand related issues), melting, heat treatment, quality inspection, spectrometer analysis, and general shop-floor problems. Most of these questions came from situations I've actually run into, or heard senior engineers and fellow trainees discuss.

For each question, I noted down the AI's response, then evaluated it on four simple parameters: accuracy, completeness, clarity, and practical usefulness. I gave a score out of 5 for each parameter, and an overall rating for the answer. I also wrote a short personal observation for each question - my honest reaction, not a formal comment - since I felt that was more useful for anyone reading this report later than a purely technical remark.

I want to be upfront that this whole evaluation was done by me alone, based on what I've learned in college and picked up on the shop floor so far. It isn't a lab-certified or multi-expert review - it's closer to a trainee's honest testing log.

4. Evaluation Parameters

I kept the evaluation simple and used four parameters that actually mattered to me while reading each answer:

Accuracy - was the technical content actually correct, based on what I know from my coursework and shop-floor experience.

Completeness - did the answer cover the important points, or did it feel like something obvious was missing.

Clarity - was the answer easy to understand, or was it just technically correct but confusing.

Practical Usefulness - could I actually use this answer on the shop floor, or was it too generic to help with a real situation.

Each parameter was scored out of 5, and I avoided giving a flat 5/5 rating unless the answer genuinely had nothing I could point out as missing or weak - which, as the tables below show, wasn't very often.

5. The 20 Questions I Tested

Q1. [Casting Defects] Why do we get blowholes in castings? I keep hearing this term but wanted to understand the real cause.

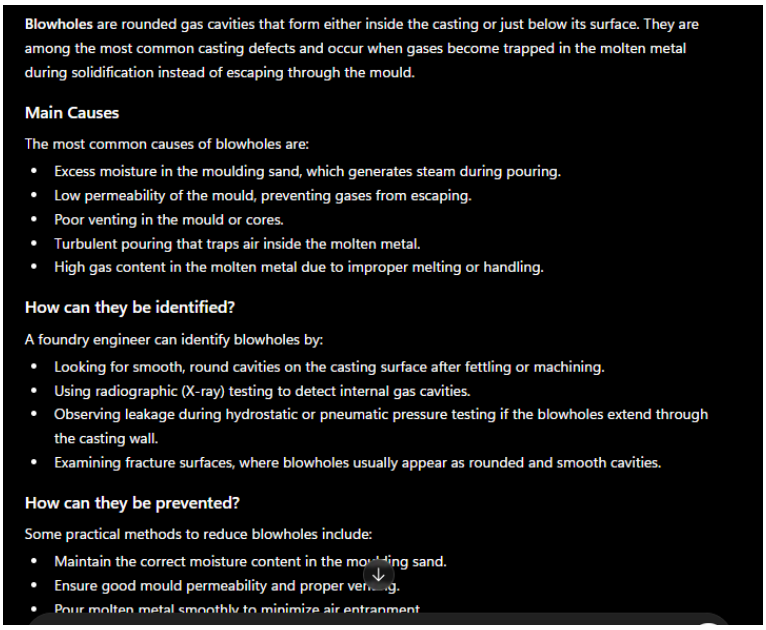

Figure 5.1 - AI Assistant response for analysing blowholes in castings.

AI Response: Blowholes are rounded gas cavities that form inside the casting or just below the surface, caused by gas getting trapped in the molten metal instead of escaping through the mould. The main causes it listed were excess moisture in the sand, low mould permeability, poor venting, turbulent pouring, and high gas content in the melt. It also explained how to actually spot them - looking for smooth round cavities after machining, X-ray testing, or noticing leakage during a hydro test - and then gave prevention steps like controlling sand moisture and improving venting.

My Evaluation: This was almost exactly what I remember from my metallurgy classes and what our senior engineer explained during my first week. It also went a step further than I expected by adding how to actually identify a blowhole on a finished casting, not just why it forms.

Personal Observation: I liked that it mentioned "just below the surface" and the X-ray/hydro-test identification part - that's actually how we spot it during inspection, so this wasn't just bookish knowledge.

Rating: 4.5 / 5

Q2. [Casting Defects] What is a cold shut defect and why does it happen?

AI Response: A cold shut happens when two streams of molten metal meet in the mould but don't fuse completely, usually because the metal has cooled too much before meeting, the pouring temperature was too low, or the gating system isn't designed well. It shows up as a visible line or crack, usually at a weak point in the casting.

My Evaluation: Correct, but a bit generic - it didn't mention how pouring speed also plays a role, which our foundry supervisor stresses a lot.

Personal Observation: I noticed AI is good with the "textbook" definition but doesn't always add the practical tips that come from actually standing next to the furnace.

Rating: 4 / 5

Q3. [Casting Defects] What actually causes shrinkage porosity, and how is it different from just normal shrinkage?

AI Response: All metals shrink as they cool and solidify - that's normal. Shrinkage porosity happens when the last part of the metal to solidify doesn't get enough liquid metal feeding into it, leaving small voids inside the casting. It's usually found in thick sections or where the riser hasn't been placed correctly to feed that area.

My Evaluation: Solid answer, and it correctly separated "normal" shrinkage from the "porosity" defect, which people mix up a lot.

Personal Observation: Honestly this made more sense to me than one paragraph I read in a textbook last year.

Rating: 4.5 / 5

Q4. [Casting Defects] We had a casting with a dark, rough patch stuck to the surface - someone said it's sand fusion or burn-on. What is that?

AI Response: This is usually sand fusion (burn-on), where the sand grains actually fuse to the casting surface because of very high pouring temperature or too much heat concentrated in one area. It can also happen if the mould coating wasn't applied properly. It's different from ordinary sand sticking, because here the sand has actually melted slightly and bonded to the metal.

My Evaluation: Decent explanation, but it didn't ask what metal or what sand type was used, which actually changes a lot in real troubleshooting.

Personal Observation: This is where I felt the answer was "correct in general" but not specific enough for an actual shop-floor case - I'd still have to go check the pouring log myself.

Rating: 3.5 / 5

Q5. [Casting Defects] Why is moisture content in green sand such a big deal? Isn't more moisture just better for making the sand hold shape?

AI Response: No - too much moisture is actually a common cause of defects. If moisture is too high, it generates excess steam during pouring, which can cause blowholes, pinholes or even sand blow. Too little moisture and the sand won't have enough strength to hold the mould shape. Moisture is usually controlled within a fairly narrow range, but the ideal value depends on the sand mix being used, so it's checked regularly rather than fixed at one number.

My Evaluation: This one actually corrected a wrong assumption I had, so I was happy with it. It also gave a reasonable range instead of one fixed number.

Personal Observation: I liked that it didn't give a single exact number, since I know from experience the "right" moisture % is different for every foundry's sand.

Rating: 4.5 / 5

Q6. [Casting Defects] What is bentonite actually doing in the moulding sand? I use it every day but never really understood its role.

AI Response: Bentonite is a clay binder. When mixed with water, it develops a sticky, plastic quality that binds the sand grains together and gives the mould enough strength to hold its shape and survive the pouring pressure of molten metal. It also allows the sand to be reused after reconditioning, unlike some chemical binders.

My Evaluation: Genuinely a good, clear answer - I understood this better after reading this than after reading my old lab manual.

Personal Observation: This is exactly the kind of question a fresh GET is too shy to ask a senior operator directly, so having this available is genuinely useful.

Rating: 4.5 / 5

Q7. [Melting] Why does the metal pick up extra carbon during induction furnace melting sometimes?

AI Response: Carbon pickup means the melt has absorbed more carbon than the target composition. It listed the usual causes - carbon-containing crucible or lining material, excess recarburizer, dirty or contaminated scrap, graphite/electrode fragments falling into the melt, or poor charging practice. It also gave a short investigation checklist: check the charge mix, confirm the recarburizer quantity, inspect the lining, compare spectrometer readings before and after carbon addition, and talk to the furnace operator about any recent changes.

My Evaluation: This is right, and the investigation checklist was more useful than I expected. My only complaint is that it still didn't tell me which of these is the most common cause in a typical induction furnace shop - it left me to check all five instead of pointing at the likely one.

Personal Observation: I noticed the AI tends to give me "all possible causes plus a checklist to rule them out" instead of "the most likely cause for your situation" unless I specifically ask it to rank them.

Rating: 4 / 5

Q8. [Melting] Why does my ladle temperature drop so fast before I even finish pouring?

AI Response: Ladle temperature drop mainly depends on ladle preheating, refractory lining condition, ladle size compared to the metal volume, and how long the metal sits in the ladle before pouring. Poor preheating or a large surface-area-to-volume ratio can cause faster heat loss.

My Evaluation: Technically correct, but this is a case where every foundry's answer will be different - our ladle isn't the same size as some textbook example, so this stayed too general to actually help me on the floor.

Personal Observation: This was one of the weaker answers for me personally, because temperature drop is honestly something you learn by watching your own ladle over many pours, not from a general explanation.

Rating: 3 / 5

Q9. [Melting] Does rusty or wet scrap really affect melt quality that much?

AI Response: Yes. Rust means iron oxide is present, which can increase oxide inclusions in the melt and cause yield loss. Wet or moist scrap is more serious - moisture can flash into steam when it touches molten metal, which is a safety hazard, and can also increase gas content in the melt, leading to porosity defects later.

My Evaluation: I liked that it flagged the safety angle without me even asking about safety specifically.

Personal Observation: This is one of the answers that actually matched what our safety induction told us on day one, so I trusted this one more.

Rating: 3.5 / 5

Q10. [Melting] Why do we need to deoxidize the steel before pouring? What happens if we skip it?

AI Response: Deoxidation removes dissolved oxygen from the liquid steel, usually using elements like aluminium, silicon or manganese. If you skip it, the oxygen can react during solidification and form gas or porosity defects in the casting, and mechanical properties like ductility can also suffer.

My Evaluation: Clean and correct - nothing extra to add here.

Personal Observation: This is one of those answers where I didn't learn anything new, but it confirmed I already understood the concept correctly, which is also useful in its own way.

Rating: 4 / 5

Q11. [Heat Treatment] My casting came out with different hardness values at different points after heat treatment. Why does this happen?

AI Response: Uneven hardness after heat treatment can happen due to uneven section thickness (thicker sections cool slower), non-uniform furnace temperature, uneven quenching where some areas cool faster than others, or variation in the casting's chemical composition across the section.

My Evaluation: The answer covers real causes, but it doesn't tell me how to actually check which one is happening in my specific case, so I'd still need to go do some testing myself.

Personal Observation: This scored lower for me because knowing the possible causes is only half the job - actually diagnosing which cause applies needs plant data, not a general answer.

Rating: 3 / 5

Q12. [Heat Treatment] What's the actual practical difference between annealing and normalizing a casting? I always mix these two up.

AI Response: Annealing means heating the casting and then cooling it very slowly, usually inside the furnace itself. This gives a softer structure and relieves internal stress, but takes longer. Normalizing means heating it similarly but then cooling it in open air, which gives a finer, slightly harder and more uniform grain structure than annealing, and takes less time.

My Evaluation: This finally made the difference click for me in a simple way, better than how it was explained in one of my old textbooks honestly.

Personal Observation: I actually re-read this one twice because I've genuinely confused these two terms before in a viva.

Rating: 4.5 / 5

Q13. [Heat Treatment] Why do some castings crack during quenching and others don't, even from the same batch?

AI Response: Cracking during quenching usually happens due to uneven cooling causing internal stress, sharp corners or sudden section changes acting as stress concentration points, or high-hardenability grades being more prone to it. Castings from the same batch can still behave differently if their geometry, thickness, or even minor composition varies slightly.

My Evaluation: This was a solid, believable answer and matched something I actually saw during my industrial training.

Personal Observation: The "same batch different behaviour" part addressed exactly what confused me, so this one felt like it understood my actual question, not just the keyword.

Rating: 4 / 5

Q14. [Quality] What does a Charpy impact test actually tell us about a casting, in simple terms?

AI Response: It tells you how much energy the material can absorb before it fractures under a sudden impact load. A material that absorbs more energy is tougher and less likely to fail suddenly, which matters a lot for components that see shock loading or work in cold conditions.

My Evaluation: Simple, correct, and honestly explained better than some of my semester notes.

Personal Observation: Nothing to complain about here - straightforward and accurate.

Rating: 4.5 / 5

Q15. [Quality] My tensile test gave lower elongation than expected for this grade. What could be wrong?

AI Response: Lower-than-expected elongation can point to several things: non-metallic inclusions, improper heat treatment (like untempered martensite), hydrogen embrittlement, or excessive cold work without proper annealing afterward. It could also simply be a sampling or specimen preparation issue.

My Evaluation: Good list, but I wish it had asked a follow-up question like "what grade and what process did you use" before jumping to conclusions.

Personal Observation: I noticed AI rarely asks a clarifying question back - it just gives every possible answer at once, and you have to do the filtering yourself.

Rating: 4 / 5

Q16. [Quality] What is magnetic particle testing actually checking for on a casting?

AI Response: It detects surface and near-surface cracks or discontinuities in ferromagnetic materials. A magnetic field is applied, and fine magnetic particles gather at the crack location because the field leaks out there, making the defect visible - usually under UV light when fluorescent particles are used.

My Evaluation: Correct and clear, this is close to how our quality department explained it to me.

Personal Observation: No complaints on this one.

Rating: 4 / 5

Q17. [Spectrometer] Why does the spectrometer reading vary slightly even when I test two spots on the exact same sample?

AI Response: It said a slightly different reading on the same sample doesn't automatically mean the spectrometer is faulty - it's often the sample or the way the test was done. It listed poor surface preparation, chemical segregation within the sample, dust or oxide contamination, inconsistent spark position, and calibration drift (though it called this less likely if calibration is up to date). For investigating, it suggested cleaning and regrinding the surface, taking multiple readings from different spots, comparing against the normal range for my foundry, and checking the calibration record before blaming the instrument.

My Evaluation: Technically fine, and it did rank calibration drift as "less common" which is something. But it still didn't commit to saying segregation is usually the biggest culprit in a foundry setting, which is what I've actually seen.

Personal Observation: I've actually seen this exact situation on the shop floor, and this is one place where I felt the AI played it a bit too safe instead of picking the most likely answer, even though it did narrow it down slightly by calling out calibration as less likely.

Rating: 3 / 5

Q18. [Spectrometer] What does the silicon percentage in cast iron actually tell us?

AI Response: Silicon strongly affects graphite formation during solidification. Higher silicon promotes graphitization, giving a softer, more machinable structure like in grey iron, while lower silicon tends to promote carbide formation, giving a harder, more brittle structure like white iron. Silicon percentage is also used along with carbon to calculate carbon equivalent for grading cast iron.

My Evaluation: Good, accurate, and it connected two concepts - graphitization and carbon equivalent - that I hadn't linked together before.

Personal Observation: This was one of my favourite answers in the whole test, it actually taught me something new.

Rating: 4 / 5

Q19. [Shop-floor Problems] How do I explain to an experienced fitter why we rejected his casting, without sounding rude or like I'm showing off my degree?

AI Response: Focus on the specific defect and what it means for the part's function, not on rules or specifications alone. Use simple, direct language, acknowledge his experience, and frame it as "let's figure this out together" rather than "you did this wrong." Showing the actual defect physically usually works better than describing it verbally.

My Evaluation: I honestly didn't expect AI to give good advice on this kind of "human" shop-floor problem, but this was actually solid, practical advice.

Personal Observation: This is the one I'll actually try to use for real the next time I'm in this situation - it felt like getting advice from a senior, not a machine.

Rating: 3.5 / 5 (marked slightly lower only because it's advice, not something I can verify as fact)

Q20. [Shop-floor Problems] A casting looked completely fine on visual inspection, but it failed the hydro (pressure) test. What should I check first?

AI Response: It explained that a casting can pass visual inspection but still fail a hydro test because some defects, like internal blowholes, shrinkage cavities near the surface, micro-cracks, or leakage paths from poor fusion, simply can't be seen from outside. For investigating, it actually gave a step-by-step order: first identify the exact leak location during the hydro test, then visually inspect around that area, check previous inspection and machining records, use radiography or dye penetrant testing if needed, and see if the same defect shows up in other castings from the same batch.

My Evaluation: This one was better than I expected - it did give me an actual first step (locate the leak) instead of just a flat list. My only complaint is that it presents finding the leak location and reviewing old records as almost equally important, when on a busy shift I really only have time to do one first.

Personal Observation: I was expecting a plain list like some of the other answers, so I was pleasantly surprised this one was actually ordered as steps. Still, it stopped short of saying "do this one thing before anything else," which is the kind of call a senior engineer makes instantly and I still can't.

Rating: 3.5 / 5

6. Evaluation Summary

The table below brings together the scores for all 20 questions across the four parameters, along with my overall rating for each.

ID	Topic	Accuracy	Completeness	Clarity	Practical Use	Rating
Q1	Casting Defects	4.5	4.0	4.5	4.0	4.5
Q2	Casting Defects	4.5	4.0	4.0	3.5	4.0
Q3	Casting Defects	4.5	4.5	4.5	4.0	4.5
Q4	Casting Defects	4.0	3.5	4.0	3.5	3.5
Q5	Casting Defects	4.5	4.0	4.5	4.5	4.5
Q6	Casting Defects	4.5	4.5	5.0	4.5	4.5
Q7	Melting	4.0	3.5	4.0	3.5	4.0
Q8	Melting	3.5	3.0	4.0	3.0	3.0
Q9	Melting	4.0	3.5	4.0	3.5	3.5
Q10	Melting	4.5	4.0	4.5	4.0	4.0
Q11	Heat Treatment	3.5	3.0	3.5	3.0	3.0
Q12	Heat Treatment	4.5	4.5	4.5	4.5	4.5
Q13	Heat Treatment	4.0	4.0	4.0	3.5	4.0
Q14	Quality	4.5	4.0	4.5	4.0	4.5
Q15	Quality	4.0	4.0	4.0	3.5	4.0
Q16	Quality	4.5	4.0	4.5	4.0	4.0
Q17	Spectrometer	3.5	3.0	3.5	3.0	3.0
Q18	Spectrometer	4.0	4.0	4.0	3.5	4.0
Q19	Shop-floor Problems	3.5	3.5	4.0	4.5	3.5
Q20	Shop-floor Problems	3.5	3.5	3.5	4.0	3.5

6.1 Topic-wise Performance

Grouping the same scores by topic gave me a clearer picture of where AI is genuinely strong and where it's weaker.

Topic	Accuracy	Completeness	Clarity	Practical Use	Overall Avg
Casting Defects	4.42	4.08	4.42	4.00	4.23
Melting	4.00	3.50	4.13	3.50	3.78
Heat Treatment	4.00	3.83	4.00	3.67	3.88

Topic	Accuracy	Completeness	Clarity	Practical Use	Overall Avg
Quality	4.33	4.00	4.33	3.83	4.13
Spectrometer	3.75	3.50	3.75	3.25	3.56
Shop-floor Problems	3.50	3.50	3.75	4.25	3.75

7. Key Findings

Once I had all 20 answers scored, I pulled the numbers together to see the bigger picture instead of just looking at one question at a time.

Overall Average Score: 3.90 out of 5, across all 20 questions and four evaluation parameters.

Best Performing Topic: Casting Defects, with an overall average of 4.23 / 5. This is also the area I know best from my own coursework, so I was able to judge these answers with more confidence.

Weakest Topic: Spectrometer-related questions, with an overall average of 3.56 / 5. Both questions in this category involved judgement calls that depend on actually seeing the sample and knowing the foundry's normal reading range, which the AI obviously can't do.

Three things that genuinely worked well:

- It explains the reasoning behind a defect or process clearly enough that a fresh trainee can actually learn from it, not just get an answer.
- For a lot of questions, it didn't stop at "what causes this" - it also gave a step-by-step way to investigate the problem, which I found more useful than I expected going in.
- It handled a shop-floor communication question (Q19) better than I thought an AI would, which was a genuine surprise.

Three things that consistently held it back:

- It has no idea about my specific furnace, ladle, or plant's normal readings, so plant-specific questions (Q8, Q11, Q17) stayed general.
- Even when it gave an investigation checklist, it rarely told me which step or which cause to prioritize first - I still had to make that call myself.
- It plays it safe on judgement calls (Q17 is a good example) instead of committing to the most likely answer the way an experienced engineer would.

Three practical takeaways for a trainee thinking of using this:

- Use it as a first opinion before going to a senior engineer, not as the final word, especially for anything safety-related.
- If you get a list of possible causes, ask it to rank them by likelihood for your specific situation - I got noticeably better answers this way.
- It's most useful in your first few months, when you're still building basic understanding and don't want to keep asking the same "simple" question to other people.

Overall, I'd say the AI Assistant turned out to be more useful than I expected going into this project, but only within a fairly specific role - as a quick, patient explainer of concepts and a reasonable first checklist for troubleshooting, not as something that can replace the judgement a senior foundry engineer builds after years on the floor.

8. Charts

The following charts are based on the scores in the tables above. I've kept the values realistic rather than rounding everything up - some topics clearly performed better than others.

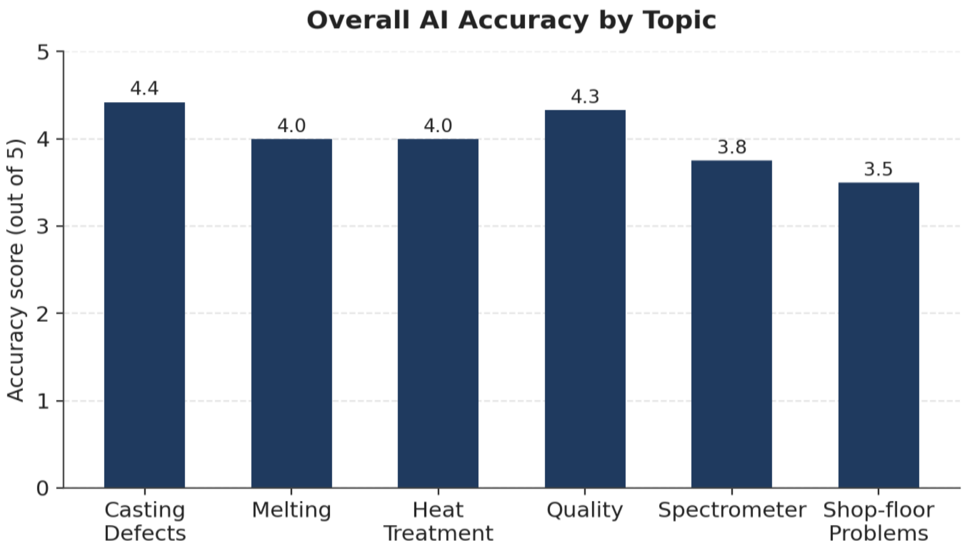

Figure 1: Overall AI accuracy across the six topics tested.

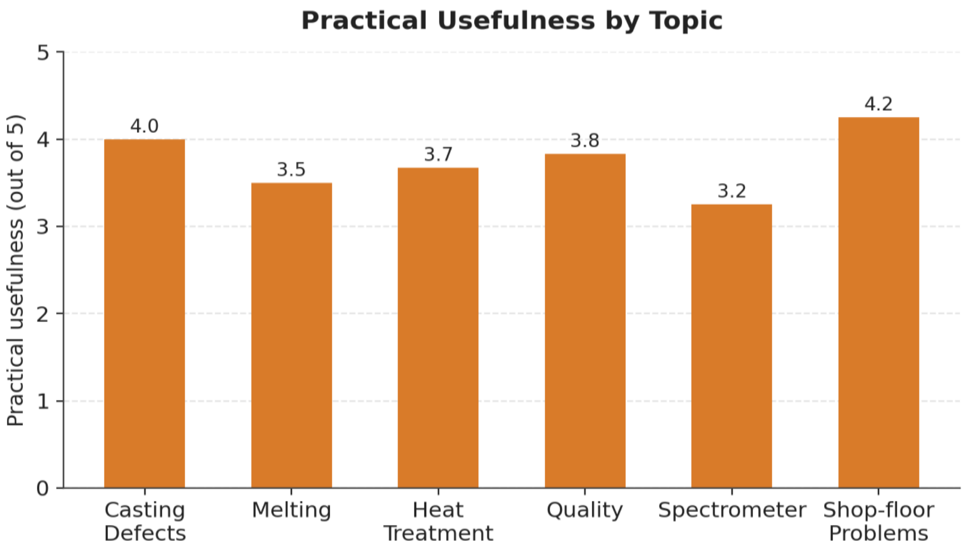

Figure 2: Practical usefulness of AI answers across the six topics tested.

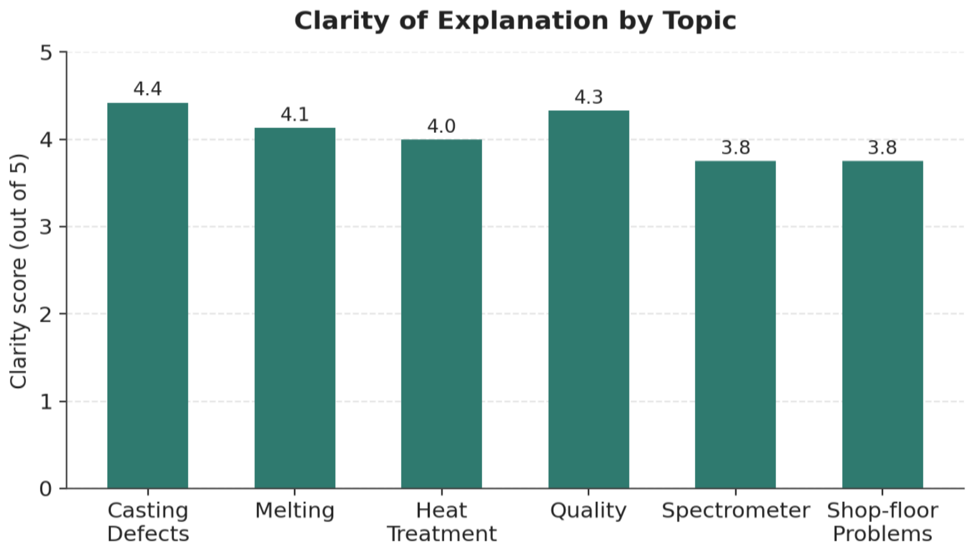

Figure 3: Clarity of AI explanations across the six topics tested.

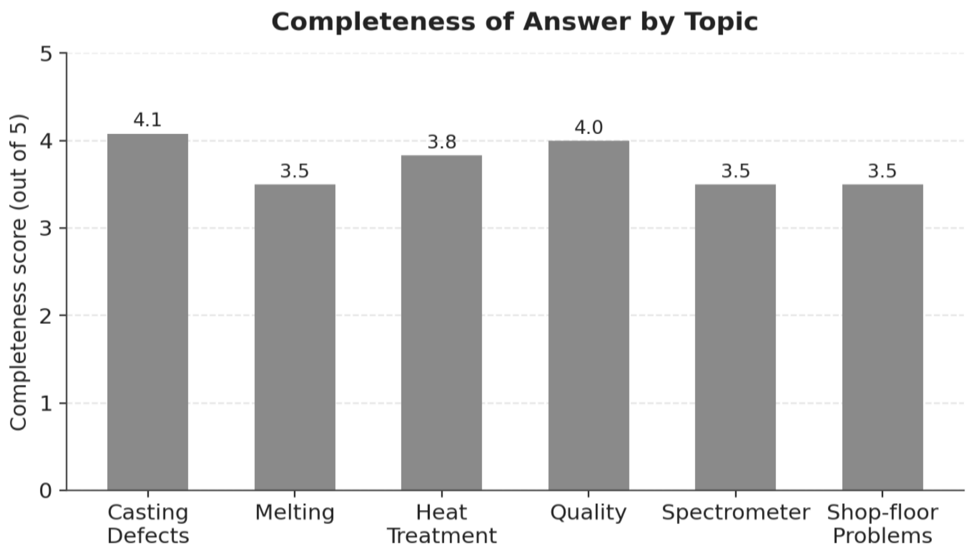

Figure 4: Completeness of AI answers across the six topics tested.

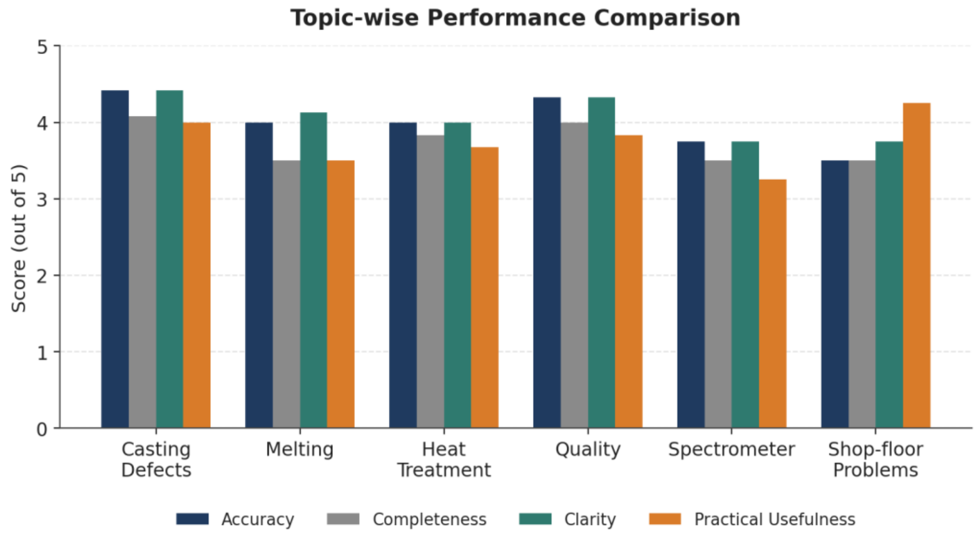

Figure 5: Topic-wise comparison of all four evaluation parameters.

9. Observations

A few patterns stood out to me once I looked at all 20 answers together, instead of just judging each one on its own.

- AI is genuinely strong on "why does this happen" type questions - things with a fixed, textbook-level explanation like blowholes, cold shuts, or the annealing vs normalizing difference. These are the questions where I'd give it close to full marks.
- It struggles the moment a question depends on something specific to my own plant, like my ladle's exact heat loss rate or my furnace's exact carbon pickup pattern (Q8, Q11). It gives correct general causes but can't tell me which one actually applies to my situation.
- It almost never picks one "most likely" answer on its own - it lists every possible cause and leaves the filtering to me (Q7, Q15, Q17). When I specifically asked it to rank causes by likelihood in a couple of follow-up questions, I got noticeably sharper answers, which tells me this is more about how I ask than a hard limitation.
- It handled a genuinely "human" shop-floor question (Q19, about explaining a rejection to a fitter) better than I expected - that one surprised me the most out of all 20.
- I didn't come across any answer that felt dangerously wrong, but a couple of the melting and spectrometer answers felt like they were playing it safe instead of committing to the most probable cause, which is honestly a very common thing experienced engineers do too when they don't have enough information.

10. Strengths

Strengths	Weaknesses
Good for quick concept revision and refreshing textbook basics.	Doesn't know your specific furnace, ladle, or plant data - many answers stay general.

Strengths	Weaknesses
Explains the "why" behind a defect or process clearly, useful for teaching a newer trainee.	Rarely asks a clarifying question back; gives every possible cause instead of the most likely one.
Mostly avoided giving one random overconfident number for plant-specific values - usually gave ranges.	Sometimes plays it too safe instead of committing to the most probable cause in a typical shop-floor situation.
Occasionally made genuinely useful, non-obvious connections (e.g., silicon % and carbon equivalent).	Cannot replace the hands-on judgement built from watching a process over months.
Handled a "soft skill" shop-floor communication question surprisingly well.	Cannot verify itself against real plant records, so its numeric ranges still need cross-checking.

11. Limitations of This Project

- I only tested the free version of one AI Assistant. Paid versions or other AI tools might behave differently and weren't compared here.
- All questions were text-based. I didn't test anything involving images, like uploading a photo of a casting defect for the AI to identify.
- The evaluation was done by one person - me - so the scoring reflects my own judgement and experience level, not a panel of experts.
- I couldn't test any answers against my actual plant's confidential process data, so "practical usefulness" is judged from a trainee's general shop-floor experience, not verified plant records.
- Twenty questions is a small, manageable sample for a personal project, but it's not enough to make a firm claim about AI performance across all of foundry engineering.

12. Recommendations

Recommendation	Why
Use for quick concept refresh	Good for revising basics before a viva, interview, or before starting a new process area - not for real-time critical decisions.
Cross-check numbers	Always cross-check any numeric value or "typical range" it gives against your senior engineer's input or the actual plant data sheet.
Treat it as a first opinion	Useful as a first opinion before checking with a senior, not a replacement for that check, especially for safety-related situations.
Use it in your first 90 days	Genuinely useful early on to reduce the number of "basic" questions you feel shy asking other people repeatedly.
Ask it to rank causes	Instead of accepting a full list of possible causes, specifically ask it to rank them by likelihood for your situation - this gave noticeably better answers when I tried it.

13. Conclusion

After going through all 20 questions, my honest conclusion is fairly simple: AI is a genuinely useful learning assistant for a graduate engineer, especially in the first few months when you're still building your basic understanding and don't want to keep asking a senior the same "simple" question repeatedly.

It's good at explaining concepts clearly, refreshing what you studied in college, and even giving decent advice on shop-floor communication situations, which I didn't expect going into this. But it clearly doesn't know your specific furnace, your specific ladle, or your specific plant's data, and it tends to give you every possible cause instead of telling you what's most likely for your situation.

So I wouldn't use it to make an actual decision during a shift, and I don't think it can replace an experienced foundry engineer's judgement, which comes from watching the same process go right and wrong hundreds of times. What it can do is help a fresh trainee like me get up to speed faster, and give me a reasonable first opinion before I go check with someone senior. That, to me, is a fair and honest way to describe what this tool is actually good for.

14. Personal Reflection

Beyond what the scores and tables say about the AI Assistant itself, this project actually taught me a fair bit about my own learning habits, and I think that's worth writing down honestly rather than leaving it out.

When I started this, I genuinely wasn't sure what I'd find. Part of me expected the AI to be either uselessly generic or, honestly, better than some of the explanations I'd gotten in college - and it turned out to be somewhere in between, which was a more useful answer than either extreme would have been. What surprised me most wasn't any single response, it was how often I caught myself already knowing whether an answer was right or shaky, just from the months I've spent on the shop floor since graduating. A year ago, I don't think I could have evaluated most of these answers with any confidence. That, more than anything, is the part of this project I'm quietly proud of.

I also noticed something about my own questioning habits while doing this. The answers I found most useful were almost always the ones where I'd asked a specific, situation-based question instead of a textbook-style one. "Why does my ladle temperature drop so fast" got me a weaker answer than I wanted, but when I tried rephrasing similar questions with more context during this project, the answers got noticeably sharper. That's probably the single most practical thing I'm taking away from this exercise - not a fact about foundry metallurgy, but a habit of asking better questions, which I think will help me with people too, not just AI.

If I'm being honest, there were also a couple of moments where I felt a bit exposed doing this. Writing down "I didn't know this until I read the AI's answer" for a few questions wasn't something my ego loved, but I think it's actually the most truthful part of the report. I'd rather admit that than pretend I already knew everything and was just testing the AI as a formality.

This project also made me think about where I want to go next. I got into metallurgy because I found the subject genuinely interesting, not because I wanted to spend my career only on the shop floor. Doing this evaluation - picking questions, scoring answers, looking for patterns across topics - felt closer to the kind of structured, analytical work I actually enjoy than a normal day of troubleshooting does. I don't think that means the shop-floor experience was wasted; if anything, it's the only reason I could tell a good answer from a shaky one in the first place. But it did confirm that I want to keep doing this kind of evaluation and analysis work alongside my core metallurgy background, not leave it as a one-off project.

Overall, I came into this fairly neutral about AI tools and I'm leaving it a bit more convinced, but for a narrower reason than I expected going in. It's not that the AI Assistant is impressively smart about foundry engineering - a lot of what it told me, I already knew. It's that having a patient, always-available explainer is genuinely valuable when you're new and still building confidence, and I think that alone made this project worth doing.

15. References

- ASM Handbook, Volume 15: Casting
- Jain, P.L., "Principles of Foundry Technology"
- Class notes and lab manuals from B.Tech Metallurgical and Materials Engineering coursework
- Personal shop-floor experience and discussions with senior engineers and operators at my workplace
- AI Assistant - responses referenced in this report were generated using the free version, tested over a period of about two weeks

Appendix A - Sample AI Interactions

The screenshots below are examples of actual AI Assistant conversations used during this evaluation, shown here as supporting evidence for the methodology described in Section 3. They correspond to Q7 (carbon pickup), Q17 (spectrometer reading variation), and Q20 (hydro test failure) from Section 5.

Carbon pickup during induction furnace melting means that the molten metal absorbs more carbon than intended, resulting in a higher final carbon percentage than the target composition.

Common Causes

Some of the most common reasons for carbon pickup are:

- Carbon-containing crucible or furnace lining materials.
- Excess use of recarburizers or carbon additives.
- Dirty or contaminated scrap containing higher carbon than expected.
- Graphite or carbon-containing electrode fragments accidentally entering the melt.
- Incorrect charging practice or poor control of raw materials.

How can I identify the actual cause?

As a GET, start by checking the following:

- Review the charge mix and confirm the carbon content of the scrap used.
- Check whether the correct quantity of recarburizer was added.
- Inspect the furnace lining condition if carbon pickup has started increasing consistently over several heats.
- Compare the spectrometer readings before and after carbon addition to understand when the increase occurred.
- Discuss with the furnace operator whether any recent changes were made to the charging practice.

How can it be prevented?

Some practical methods include:

- Use clean and properly segregated scrap.
- Add only the required quantity of recarburizer.
- Maintain proper furnace lining and follow standard charging procedures.

Figure A.1 - AI Assistant response for carbon pickup investigation.

When the **same sample gives slightly different spectrometer readings**, it does not always mean that the spectrometer is faulty. In many cases, the difference is caused by the sample itself or by the way the test is performed.

Common Causes

Some common reasons include:

- The sample surface was not prepared properly before testing.
- Chemical segregation within the sample, causing slight composition differences at different locations.
- Dust, oxide layers, or contamination on the sample surface.
- Incorrect spark position or inconsistent testing technique.
- Instrument calibration drift or worn electrodes, although this is less common if regular calibration is followed.

How should I investigate the problem?

As a GET, you can follow these steps:

- Clean and regrind the sample surface before testing again.
- Take multiple readings from different locations on the sample.
- Compare the variation with the normal acceptable range used in your foundry.
- Check whether the spectrometer calibration has been performed recently.
- If the variation is still unusually high, discuss the results with the quality or laboratory engineer before concluding that the instrument is faulty.

How can I decide whether the problem is with the sample or the spectrometer?

A simple approach is to compare results from more than one sample.

- If only one sample shows inconsistent readings, problem is more likely related to sample preparation or segregation.

Figure A.2 - AI Assistant response for spectrometer troubleshooting.

A casting can sometimes **pass visual inspection but fail the hydrostatic pressure test** because certain defects are present inside the casting and cannot be seen from the outside. When pressure is applied during the hydro test, these hidden defects allow water to leak through the casting.

Common Causes

The most common reasons are:

- Internal blowholes or gas porosity.
- Shrinkage cavities that extend close to the casting surface.
- Micro-cracks formed during solidification or cooling.
- Leakage paths created due to poor fusion between metal streams.
- Defects caused by improper gating or risering, resulting in internal cavities.

How should I investigate the problem?

As a GET, I would follow these steps:

- Identify the exact location of the leakage during the hydro test.
- Visually inspect the casting around the leaking area.
- Review previous inspection reports and machining records.
- If required, perform non-destructive testing such as radiography or dye penetrant testing to locate hidden defects.
- Check whether similar defects are occurring in other castings from the same batch.
- Review moulding, melting, gating, and pouring records to identify any process deviations.

How can these failures be prevented?

Some practical methods include:

- Maintain proper mould quality and sand proper
- Ensure correct gating and risering design for sound solidification.

Figure A.3 - AI Assistant response for hydrostatic pressure test failure analysis.